

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457406.6971 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.083 +/- 0.045
Transit depth (Rp/Rs)^2	0.7 +/- 0.75 [%]
Orbital Inclination (inc)	85.2 +/- 2.81
Airmass coefficient 1 (a1)	21.471 +/- 0.032
Airmass coefficient 2 (a2)	-2.68 +/- 1.31
Scatter in the residuals of the lightcurve fit is	0.43 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	4.75
Transit Duration (day)	0.0611 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.083 ± 0.045 vs 0.1178 (deviation 0.77σ)
Duration fit vs published	0.0611 d vs 0.125 d (deviation 30.43σ)
Mid-time O-C	0.2 min
Depth SNR	1.8
Residual scatter	0.43 %

Light curve

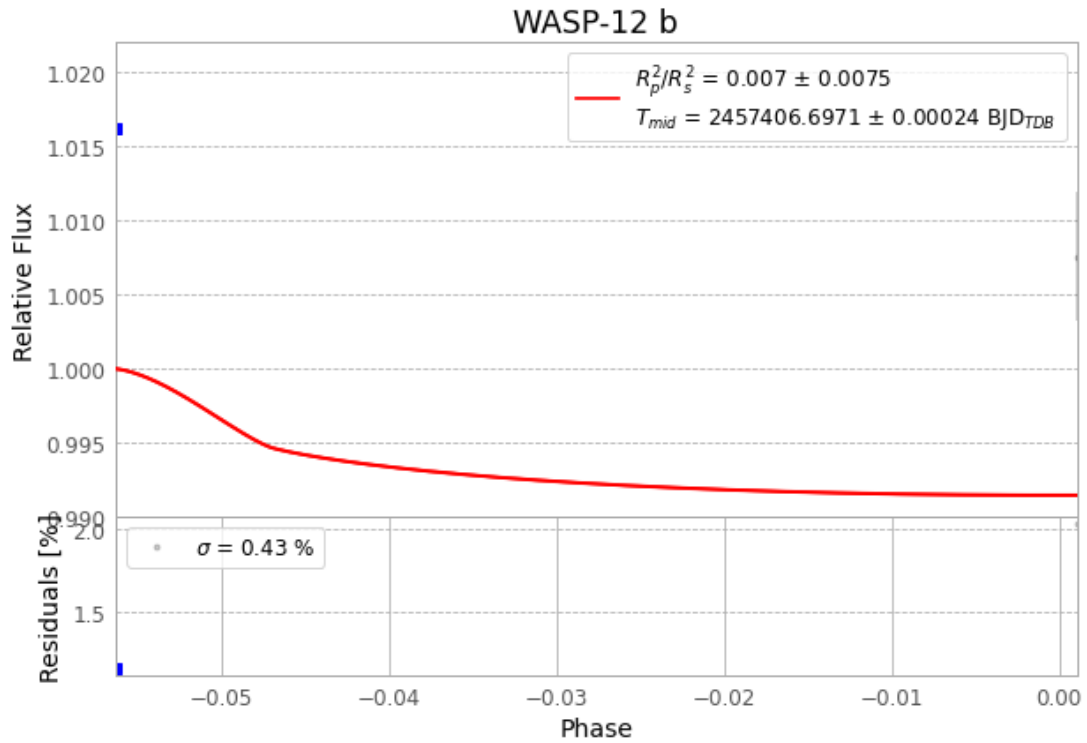


Figure 1 — FinalLightCurve_WASP-12 b_2016-01-18.png (SHA-256 ecee100cc4f2af11...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [460, 119], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9964af670ef7f40c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12160119022112.FITS → WASP-12160119072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160119.log (ab3c568915cc...), FinalLightCurve_WASP-12 b_2016-01-18.png (ecee100cc4f2...), FinalLightCurve_WASP-12 b_2016-01-18.csv (054674fe5493...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:12Z.